

Waylet: Self-Contained Haptic Device for Park-Scale Interactions

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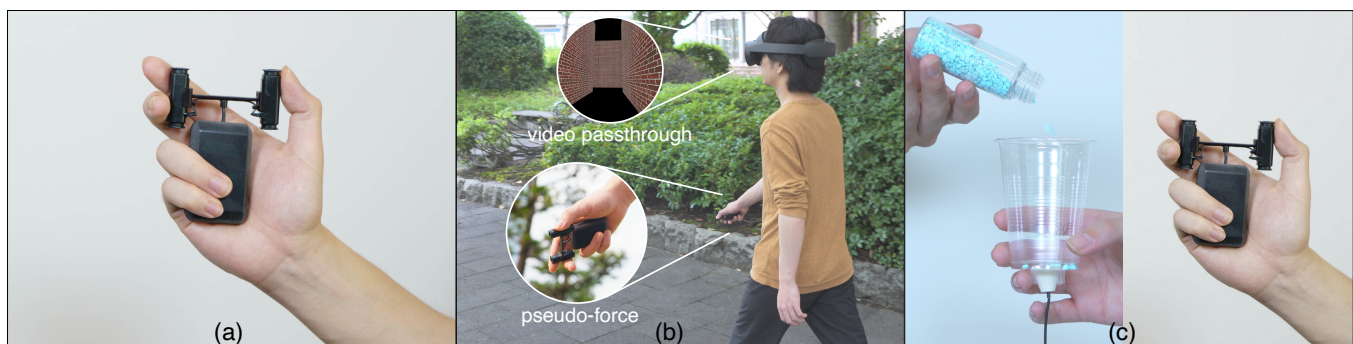


Figure 1: Waylet is a self-contained haptic device that provides translational and rotational pseudo-forces by asymmetric vibration and vibrotactile textures for park-scale interactions. (a) Waylet prototype. (b) Haptic navigation system in a park-scale mixed reality environment. (c) Haptic rendering demonstration using pseudo-forces and vibrotactile textures.

ABSTRACT

Although various haptic devices have been proposed so far, most of them are limited to use in laboratory or indoor environments, because such haptic devices are wired to external equipment for power supply and control. This can be a critical issue, particularly when using them in freely moving situations. In this study, we proposed a self-contained haptic device called Waylet for park-scale interactions. Waylets can provide translational and rotational pseudo-forces via asymmetric vibrations. To demonstrate the feasibility of our concept, we demonstrate an intuitive haptic navigation system in a park-scale mixed-reality environment with haptic rendering.

CCS CONCEPTS

• **Human-centered computing** → **Haptic devices**; *Mobile devices*.

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KEYWORDS

haptics, park-scale interaction, pseudo-force

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1 INTRODUCTION

Although various haptic devices have been proposed, most are limited to use in laboratory or indoor environments. This is because these haptic devices are wired to external equipment for power supply and control. This can be a critical issue, particularly when using them in freely moving situations. Therefore, most haptic devices have been validated only in laboratory settings, such as haptic instructions [Kappers et al. 2022]. Self-contained haptic devices are essential for expanding the applications of haptic devices to realistic environments, such as outdoor navigation.

In this study, we propose a self-contained haptic device called Waylet for park-scale interactions. Waylet is a hand-held device consisting of two vibrotactile actuators and a control unit (Figure 1a). The combination of actuators can provide translational and rotational pseudo-forces [Culbertson et al. 2017; Tanabe et al. 2020],

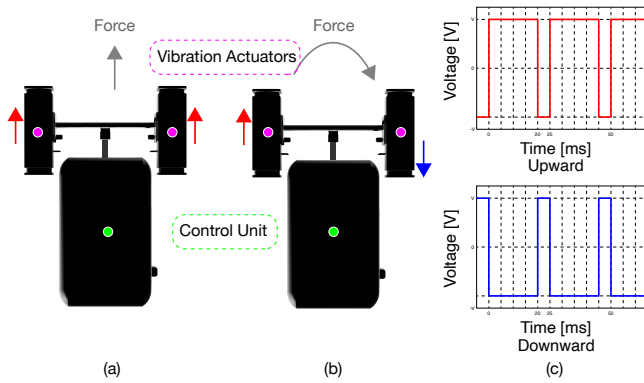


Figure 2: Principles of (a) translational and (b) rotational pseudo-forces and (c) asymmetric vibrations.

along with simple vibration signals on the user's fingers (Figures 2). The control unit includes a drive circuit, a battery, and a wireless communication module to receive external triggers, such as mobile phone signals. These features enabled haptic interactions to occur in realistic environments. As an example, we demonstrate a haptic navigation scenario using illusory traction (Figure 1b). Although a similar scenario was previously proposed by Rekimoto [Rekimoto 2013] and Amemiya et al. [Amemiya and Gomi 2014], our device establishes a more intuitive navigation using translational (forward/backward) and rotational (left/right turn) pseudo-forces.

2 DESIGN AND IMPLEMENTATIONS

2.1 Haptic Device

Waylet weighed approximately 60 g and consisted of two vibrotactile actuators and a control unit including a circuit, battery, and wireless communication module (Figure 2a, b). The housing was designed to be compact enough for convenience using a single hand. Waylet was controlled using Bluetooth or WiFi. The device was operated continuously for approximately 60 min at full charge.

The vibrotactile actuator was a compact and lightweight voice coil actuator (HapCoil-One, Tactile Labs) capable of providing high-output vibrotactile stimuli. To avoid the generation of unwanted torque sensations, the two vibrotactile actuators were fixed in parallel to a thin flat plate.

The control unit mainly consisted of a microcontroller (ESP32, Espressif Systems Pte. Ltd.), wireless communication module, digital audio amplifier (NS4168, Shenzhenshi YONGFUKANG Technology co., LTD.), and battery (Data Power Technology, 500 mAh). A digital-analog converter (DAC) in the microcontroller generated stimulation signals.

2.2 Haptic Feedback

Translational and rotational pseudo-forces are induced by controlling the direction of the asymmetric vibration input to the two actuators (Figure 2). When subjected to an upward asymmetric vibration, the actuators generated a forward pseudo-force sensation (Figure 2a). Conversely, in response to a downward asymmetric vibration, they generated a backward pseudo-force sensation. Waylet

can provide rotational pseudo-forces by changing the combination of the force directions exerted by each vibrotactile actuator (Figure 2b). Preliminary investigations revealed that the asymmetric vibration used a 40 Hz square wave with a duty ratio of 0.25 (Figure 2c).

To render the surface vibrations of an object into haptic sensation, we used acoustic vibration proposed by Minamizawa et al. [Minamizawa et al. 2012]. This method uses a contact microphone to capture the surface vibrations and subsequently feeds the acquired signals to the vibrotactile actuators, thus rendering the vibrotactile textures.

3 USER EXPERIENCES

In SIGGRAPH Asia 2023, the attendees can experience an intuitive haptic navigation system in a park-scale mixed-reality environment. Specifically, each attendee, equipped with a Waylet device and head-mounted display (Meta Quest Pro, Meta) *actually*, can walk in a virtual city as illusory traction guides them toward the goal (Figure 1b). During this navigation scenario, the attendees can experience haptic warning notifications, alerting them to potential collisions with other road users (e.g., cars). To further underscore the versatility of Waylet, we plan to demonstrate haptic rendering. The participants can tangibly sense the haptic qualities of the materials (pebbles or rubber balls) being poured into a cup in a virtual environment through pseudo-forces generated by vibrotactile textures [Minamizawa et al. 2012] (Figure 1c).

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